DUET: Dual-Granularity Address Translation for High-Capacity SSDs

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***Abstract*—High-capacity QLC SSDs cannot keep a 4 KB page-level mapping table in on-device DRAM. Existing FTLs that fit a limited mapping DRAM budget slow reads with mapping misses, slow writes with block-level merges, or turn every small write into a read-modify-write (RMW) of a *uniform coarse indirection unit* (uniform coarse IU). We present DUET, an FTL that keeps the coarse IU and maps writes smaller than a coarse IU as 4 KB fine pages in a fixed-size DRAM hash table. When the table fills, DUET spills the fine-map entries of a coarse IU to NAND instead of moving data, and it merges fine pages one coarse IU at a time. In NVMeVirt with 1/8 of the page-level mapping DRAM, DUET reaches 93% of the throughput of a page-level FTL with unlimited DRAM, whereas on-demand mapping, block-level hybrid mapping, and a uniform coarse IU reach at most 73%, 57%, and 66%, respectively.**

I. Introduction

The flash translation layer (FTL) is the SSD firmware layer that maps every logical page to its physical location. Because NAND flash cannot be overwritten without erasing a whole block, physical pages are programmed sequentially within a block and every update is written out of place, so the FTL maintains a mapping table that translates each logical page to its current physical page. Most commercial SSDs deployed today map at 4 KB granularity and keep this table in on-device DRAM.

With QLC NAND, SSDs of 64 and 128 TB are being released and developed [1]. At 4 KB granularity, a 64 TB SSD has 2^{34} logical pages, so each entry needs at least 34 bits and the table takes 68 GB. A 128 TB SSD needs 35-bit entries and 140 GB. Providing this much on-device DRAM is impractical because of its cost, power, and board area [2], so large SSDs have to translate addresses with a mapping DRAM budget that is only a fraction of the full page-level table (Figure 1a).

Techniques that reduce the mapping DRAM fall into two broad classes, and both pay for it in latency. *On-demand mapping* [3], [4] (Figure 1b) keeps the page-level table in flash and keeps part of it in an on-device map cache. A read that misses in the map cache must first read its mapping entry from NAND before it can read the data. This extra read is synchronous, so read latency spikes whenever the map cache misses. *Block-level hybrid mapping* [5]–[8] (Figure 1c), on the other hand, maps most data per NAND block and appends updates to page-mapped log blocks. When free space runs low, the log blocks are merged back into their data blocks, and a merge must copy all valid pages of a whole NAND block. Each 3D NAND generation enlarges the NAND block [9], so a merge copies more pages, and these copies compete with host writes for NAND bandwidth. In our measurements, the p99 write latency of FAST [6], a representative block-level hybrid mapping FTL, is 2.8–3.4× that of a page-level FTL under random I/O.

Many recent commercial high-capacity SSDs instead map all data with a *uniform coarse indirection unit* (uniform coarse IU) of 16–64 KB [10]. The whole coarse table fits in on-device DRAM, with no mapping misses and no block-level merges. The cost moves to small writes. When a write is smaller than a coarse IU, the FTL reads the rest of the unit, merges the new data, and writes the whole unit again

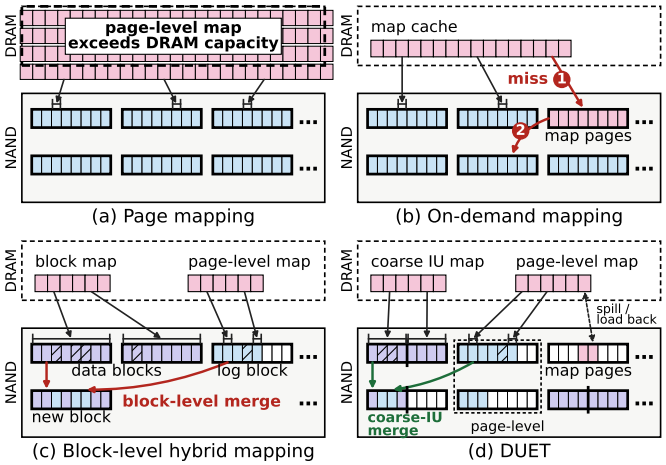


Fig. 1. Address translation of four FTLs. In (d), page-level mappings spill to map pages instead of being merged when the DRAM table is full, and are loaded back on access.

(read-modify-write, RMW). This raises the write amplification (WA) and lengthens write latency.

We present **DUET** (**DU**al-granularity **E**conomical **T**ranslation), an FTL that keeps two mapping granularities, the coarse IU and the 4 KB page, under a fixed mapping DRAM budget (Figure 1d). Most data are mapped per coarse IU, and recent writes smaller than a coarse IU are mapped at 4 KB granularity. They are written out of place as 4 KB pages, and the corresponding pages of the coarse IU become invalid, so the host write path has no RMW. Most of the page-level mappings stay in on-device DRAM, and DUET uses NAND only to stage a small part of them so that they fit the DRAM budget, which is fixed in hardware. DUET thus needs far fewer mapping reads than on-demand mapping, which reduces its read latency. When either the page-level or the coarse-IU-level area runs out of space, its garbage collection (GC) merges the 4 KB pages of a coarse IU into it and rewrites the whole coarse IU. A merge rewrites only one coarse IU regardless of the NAND block size, so DUET never performs the block-level merges of existing hybrid mapping FTLs.

We implement DUET and six other FTLs in NVMeVirt [11], a kernel-level NVMe SSD emulator, and evaluate them under a limited mapping-DRAM budget with microbenchmarks, application workloads, and real-world traces. DUET reaches 93% of the throughput of a page-level FTL with unlimited DRAM, whereas on-demand mapping, block-level hybrid mapping, and a uniform coarse IU reach at most 73%, 57%, and 66%, respectively. Its average read latency is up to 2.4× lower than that of on-demand mapping, and its p99 write latency is 4.6× lower than that of block-level hybrid mapping on TPC-C.

This paper makes the following contributions:

- An analysis showing that existing FTLs fit a limited mapping DRAM budget only by paying in read latency (on-demand map-

ping), merge cost and tail latency (block-level hybrid mapping), or RMW (uniform coarse IU).
- The design of DUET, which combines dual-granularity mapping with coarse-IU-level merging.
- An implementation of DUET and six other FTLs in a common emulator. Under the same DRAM budget, DUET achieves the highest average throughput among the bounded-DRAM FTLs, within 7% of a page-level FTL with unlimited DRAM.

II. Background and Motivation

A. Address Translation under a DRAM Budget

NAND flash is read and programmed in pages (4–16 KB) but erased only in much larger blocks, and a page cannot be overwritten before its block is erased. The SSD controller therefore writes every update out of place and runs a flash translation layer (FTL) that maps each logical page to its current physical page. Old copies become invalid, and garbage collection (GC) reclaims a block by copying its remaining valid pages elsewhere and erasing it. These copies make the number of pages written to NAND flash larger than the amount of host writes. The ratio is the write amplification (WA), and spare over-provisioned (OP) capacity keeps it bounded.

A page-level FTL keeps one entry per 4 KB logical page, so its table is about 0.1% of the capacity and grows linearly with it. Conventional commercial SSDs keep the whole table in on-device DRAM, but high-capacity QLC SSDs can hold only a fraction of it because of the cost, power, and board area that so much DRAM would take [2]. The FTL must then either keep part of the mapping in flash or map data in larger units. The three approaches below take these two routes, and each one moves the cost to a different part of the I/O path.

On-demand mapping. On-demand mapping stores the page-level table in non-volatile NAND flash as flash translation pages (FTPs) and keeps only recently used entries in an on-device *map cache*, as introduced by DFTL [3]. A small global translation directory (GTD) locates each FTP. A read that misses in the map cache must first read the FTP and only then the data page, so two flash reads are serialized on the read path. Evicting a dirty entry also writes its FTP back, which adds write amplification. LeaFTL [4] compresses the map cache into learned linear segments, so the same DRAM covers more pages. The segments, however, are built only from runs of sequentially placed pages, and a mispredicted location imposes an extra flash read. For randomly written data, both FTLs miss often, and every miss turns one read into two dependent flash reads. Read latency therefore depends on the map-cache hit ratio, not on the device.

Block-level hybrid mapping. Block-level hybrid FTLs such as BAST (block-associative sector translation) [5], FAST (fully-associative sector translation) [6], and Janus-FTL [7] map most data per NAND block, which needs only one entry per block, and absorb updates in a small pool of page-mapped *log blocks*. When the pool runs out, a log block is reclaimed by *merging*. The valid pages of a logical block are copied from its data block and the log block into a free block, and the old data block and log block are erased. A merge imposes a higher overhead than the GC of a page-level FTL. Page-level GC picks the block with the fewest valid pages, so it copies few pages per reclaimed block. A merge instead rewrites the whole data block even when only a few of its pages were updated, and one log block can hold updates to many logical blocks, all of which must be merged before it is erased.

The merge copies compete with host writes for NAND bandwidth, which raises write latency and lowers write throughput. Each merge also rewrites a whole NAND block, and NAND blocks keep growing with 3D NAND generations [9], so the work behind a single merge grows with them. CoMap [12] selects page or block mapping for each write request with reinforcement learning, but the data that it maps per block are still reorganized at NAND-block granularity.

Uniform coarse-IU mapping. High-capacity SSDs in production avoid both problems by mapping all data with a *uniform coarse indirection unit* (uniform coarse IU) of 16–64 KB [10]. One entry per coarse IU shrinks the table by the coarse-IU size over 4 KB, so the whole table fits in DRAM: every read takes a single flash access, and there are no block-level merges. The cost moves to writes smaller than a coarse IU. The FTL must read the rest of the unit, merge the new data, and program the whole unit again (read-modify-write, RMW). The extra reads lengthen the write path, and the rewritten pages raise write amplification by up to the coarse-IU size over the write size. In our evaluation the 1/8 budget allows a 32 KB coarse IU (4 B per 32 KB), which raises WA by 2.7–3.3× over a page-level FTL on YCSB-A and YCSB-B.

B. Motivation

Figure 2 compares the three approaches with a page-level FTL that keeps its whole table in DRAM. All FTLs run in NVMeVirt [11] with the same NAND timing and the same 1/8 mapping-DRAM budget. Each bar is the geometric mean over the 10 workloads of Section IV (real traces, applications, and raw I/O), normalized to the page-level FTL.

On-demand mapping slows reads. With mapping misses, DFTL needs 1.46 flash reads per host read and LeaFTL 1.22 (Figure 2a, where LeaFTL's misprediction reads are not counted, so its value is a lower bound). Their average read latency is 2.4× and 1.7× that of the page-level FTL (Figure 2b), although the flash itself is the same.

Block-level hybrid mapping and uniform coarse IUs slow writes. FAST's block-level merges raise WA by 2.1× (Figure 2c), so its average write latency is 2.4× that of the page-level FTL (Figure 2d) and it delivers only 57% of the page-level throughput. The uniform coarse IU avoids merges, but every 4 KB write becomes an RMW of its 32 KB unit: WA grows by 2.2× and write latency by 2.8×.

DUET, shown for reference, uses the same DRAM budget as the other bounded FTLs and stays within 1.05× of the page-level FTL in flash reads per read, average read and write latency, and WA.

Goal. An FTL that fits a limited DRAM budget should therefore (i) minimize mapping reads, so that a host read costs a single flash read as with a uniform coarse IU, (ii) eliminate block-level merges, and (iii) absorb small writes without RMW. DUET meets all three by keeping the coarse IU and adding a fine-grained level within the same DRAM budget.

III. DUET Design

DUET maps every logical page at two mapping granularities (Figure 3). A *coarse map* keeps one entry per coarse IU, and a *fine map* maps the 4 KB fine pages written for parts of a coarse IU. The corresponding pages of the coarse IU, which are mapped by the coarse map, then become invalid. The coarse map is always in DRAM, so most reads need no flash read to find their mapping. Part of the fine map can move to NAND, but unlike demand-paging FTLs [3], [4], [13], most of the map stays in DRAM, so far fewer reads need a second flash read. Unlike hybrid FTLs [5], [6], which merge one block at a time, DUET merges fine pages one coarse IU at a time, which reduces the cost of each merge. Unlike coarse-IU FTLs [10], DUET writes small writes as fine pages instead of RMW.

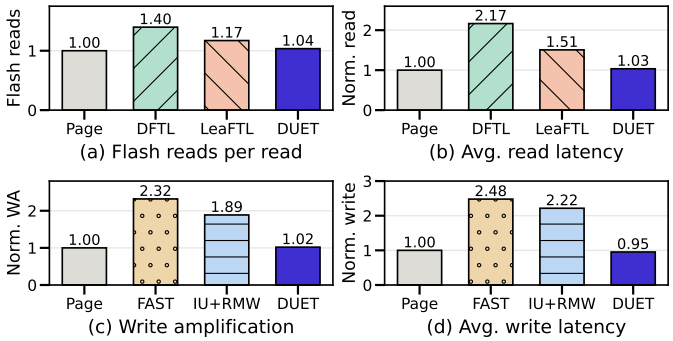


Fig. 2. Existing FTLs with 1/8 of the page-level mapping DRAM, geometric mean over 10 workloads. Latency is normalized to Page. (a) Flash reads per host read. (b) Average read latency. (c) Write amplification. (d) Average write latency.

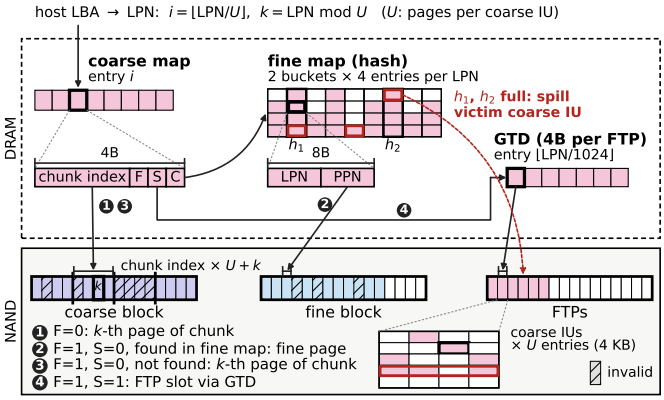


Fig. 3. DUET mapping structures.

A. Two Mapping Granularities

Coarse map. The logical space is divided into coarse IUs of U pages ($4U$ KB), so a logical page number (LPN) belongs to coarse IU $i = \lfloor \mathrm{LPN}/U \rfloor$ at page offset $k = \mathrm{LPN} \bmod U$. Each coarse IU is stored as a *chunk* of U physically consecutive, U-page-aligned pages in a *coarse block*, and its 4 B coarse entry holds the chunk index. Since a coarse entry is as large as a page-level entry, the coarse map takes $1/U$ of the DRAM of a page-level table.

Fine map. A write smaller than a coarse IU is written out of place as 4 KB *fine pages* into separate *fine blocks*. The fine map is a two-choice bucketized hash table [14]. With two candidate buckets, a bucket fills up much later than in an ordinary hash table with a single candidate bucket [14], so DUET evicts entries less often. Each 8 B entry holds an LPN and the physical page number (PPN) of its fine page. Each LPN has two candidate buckets of four entries each, and a new entry goes to the less loaded one. The table takes all of the DRAM budget that the coarse map leaves.

The table is similar to cuckoo hashing [15] in that each entry has two candidate buckets. Unlike cuckoo hashing, DUET never moves an existing entry to its other bucket to make room, so an insert does not trigger a chain of moves. When both candidate buckets are full, DUET instead writes the fine-map entries of a victim coarse IU to a *fine translation page* (FTP) in NAND and frees them from the table (*spill*, Section III-D). A table lookup therefore checks at most the eight entries of the two candidate buckets, and an insert writes into a free entry of one of them. A spilled entry has a fixed location in NAND, so one map-page read returns it.

Flag bits. Each coarse entry also has three flag bits. The *has-fine-pages* (F) bit tells whether the coarse IU has fine pages, the *spilled* (S) bit tells whether their fine-map entries are in NAND, and the *clean* (C) bit tells whether the entries in the table are identical to their copy in NAND (Section III-D).

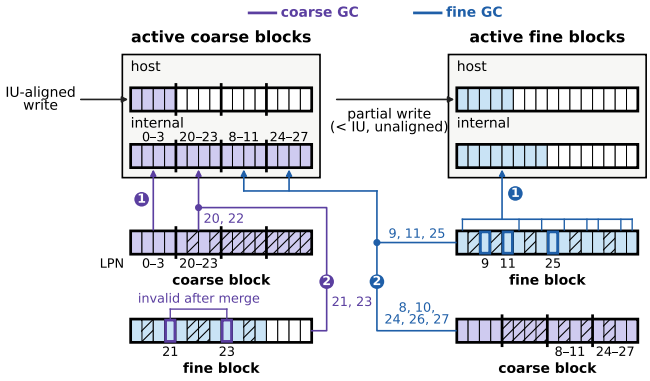


Fig. 4. DUET write and reclamation paths. Coarse GC (purple) and fine GC (blue) either (1) copy valid data or (2) merge it into new chunks.

B. Host I/O Path

Read path. A read first checks the coarse entry of coarse IU i. If F is set and S is clear, DUET looks up the LPN in the fine map. If S is set, the global translation directory (GTD) [3] locates the FTP of the LPN, and DUET reads its slot. When the fine map or the slot points to a fine page, DUET reads the fine page. Otherwise, including when F is clear, it reads page k of the chunk. A read thus reads at most three DRAM locations, the coarse entry and the two candidate buckets. A 32 B bucket lies within one DRAM row, so each location costs one random DRAM access [16]. Only a read of a spilled coarse IU adds one map-page read.

Write path. Host writes go to two *active blocks*, an active coarse block for chunks and an active fine block for fine pages. A host write is split at coarse-IU boundaries. Each IU-aligned write is written as a new chunk in the active coarse block. The old chunk and all fine pages of the coarse IU become invalid, and the corresponding fine-map entries are removed. If the coarse IU already has fine pages, DUET instead writes the IU-aligned write as fine pages while the fine blocks are below their cap (Section III-C).

A *partial write*, which is smaller than a coarse IU, is written as fine pages into the active fine block. For each fine page, DUET inserts its LPN and PPN into one of the two candidate buckets of the LPN, the less loaded one. The page of the chunk with the same LPN becomes invalid. If the LPN already has a fine page, DUET updates the PPN of its fine-map entry instead, and the old fine page becomes invalid.

DUET keeps fine pages and chunks in different blocks. Fine pages hold recently rewritten data and chunks hold colder data, and mixing data of different update frequencies in one NAND block raises the GC cost [17]–[19]. As in other FTLs [20], DUET also separates host writes from internal writes (GC and merges), so it keeps four active blocks (Figure 4). When an active block is full, DUET takes a free block as the new active block.

C. Reclaiming Path

DUET reclaims a block when the free blocks run low. Its written blocks are divided into fine blocks, which hold fine pages, and coarse blocks, which hold chunks. DUET thus has two kinds of GC. *Fine GC* reclaims a fine block, and *coarse GC* reclaims a coarse block. To find the valid data of a victim, DUET keeps a valid bitmap with one bit per chunk slot of the coarse blocks and one bit per page of the fine blocks.

Victim selection. DUET picks a victim with fewer valid pages to copy, whether it is a coarse block or a fine block. The number of fine blocks, however, is limited by the OP space. A fine page makes one page of a chunk invalid, but the chunk still takes its place in a coarse block. The coarse blocks must therefore be able to cover the whole logical capacity, and the fine blocks can only use the space beyond it. When the fine blocks reach this cap, DUET thus reclaims a fine block. Otherwise, it compares the coarse and fine blocks and runs GC on the side whose block has fewer valid pages.

Coarse GC. Coarse GC handles each valid chunk of the victim in one of two ways (purple lines in Figure 4). A chunk whose coarse IU has no fine pages (LPN 0–3) is copied unchanged to a new chunk (1). A chunk whose coarse IU has fine pages (LPN 20–23) is only partly valid, so DUET merges it (2). DUET looks up the LPNs of the coarse IU in the fine map. An LPN found there has a fine page (LPN 21 and 23), and an LPN not found there is still valid in the chunk (LPN 20 and 22). DUET reads these pages and writes them as one merged chunk. The fine pages then become invalid, their fine-map entries are removed, and the F bit of the coarse IU is cleared.

Fine GC. Fine GC handles each coarse IU with valid fine pages in the victim in one of two ways (blue lines in Figure 4). DUET either copies the fine pages to another fine block (1) or merges the coarse IU with its chunk (2). In Figure 4, LPN 9 and 11 are merged with the chunk of LPN 8–11, and LPN 25 with the chunk of LPN 24–27. Let d be the number of fine pages of the coarse IU in all fine blocks, not only in the victim. A merge reads the chunk and these d fine pages and writes one new chunk. All d fine pages then become invalid, and their fine-map entries are removed.

Copy-or-merge decision. DUET chooses one of the two ways by a cost-benefit model. The cost is the number of pages written, and the benefit is the number of pages freed in the fine blocks. Let P be the number of pages per block, v the number of valid pages in the victim, and U the number of pages in a coarse IU. For the same victim, copying costs v page writes and frees $P - v$ pages. A merge of a coarse IU with d fine pages instead costs U page writes, one coarse IU, and frees its d fine pages. The cost-benefit value, that is, the number of pages freed in the fine blocks per page written, is thus $(P-v)/v$ for copying (1) and d/U for a merge (2). DUET copies the fine pages when $(P - v)/v \geq d/U$, and otherwise merges the coarse IU. DUET makes this decision for every coarse IU that has valid fine pages in the victim fine block.

Copying fine pages keeps them in the fine blocks, whereas a merge moves them into a chunk and frees their fine-map entries. The size of the fine map thus adapts to the workload (Section IV).

D. Fine-Map Spill and Load

Under a fixed DRAM budget, the table cannot index every fine page that the fine blocks can hold. For example, a budget of 128 MB per TB of capacity, 1/8 of the page-level table, holds the whole map of an SSD with uniform 32 KB coarse IUs. If DUET uses 64 KB chunks under this budget, the coarse map takes 64 MB per TB. Since each fine-map entry takes 8 B, the remaining 64 MB holds about 8M entries, which map 32 GB of fine pages. If the fine blocks use all of the 7% OP space, however, they can hold 70 GB of fine pages per TB. The table thus has no room for more than half of their fine-map entries. The DRAM capacity is fixed when the SSD hardware is designed. Unlike the FTL, which can be updated with the firmware, it cannot be changed easily.

Another challenge is hash collisions within the fine map. An insert can find both of its candidate buckets full even when other buckets still have free slots. Hashing schemes that place a colliding entry in another bucket [15], [21]–[23] make a lookup probe more buckets or an insert move existing entries, so they need more DRAM accesses. On the other hand, merging the coarse IU whose entries are pushed out of the full buckets into the coarse map causes extra NAND writes.

DUET instead keeps the fine-map entries of some coarse IUs in FTPs in NAND, so the fine blocks can use all of the OP space without more DRAM. The fine-map entries of a coarse IU move to its FTP when the table fills (*spill*) and back to the table when the coarse IU is accessed again (*load*), and no data moves in either direction. In theory, the spilled entries map at most 7% of the data, the size of the OP space, because most data are mapped by the coarse map, which is always in DRAM.

Spill. As shown in Figure 3, the fine-map entry of each LPN has a fixed 4 B slot in an FTP, which holds the entries of 1024 consecutive LPNs. As in DFTL, the GTD locates each FTP with one 4 B entry, charged to the same DRAM budget. When an insert finds both candidate buckets full, DUET picks a victim among the coarse IUs that have an entry in the two buckets. It writes all entries of the victim to its FTP, sets its S bit, and frees its table entries. The other coarse IUs of the same FTP are spilled with the same program.

Load. An access to a spilled coarse IU reads its FTP and loads only the entries of that coarse IU into the table. The other coarse IUs of the FTP stay spilled. A write loads the entries and then updates them. A read reads the FTP before the data, loads the entries, and sets the C bit of the coarse IU. The coarse IU stays clean until one of its entries changes.

Spill victim. DUET picks a clean coarse IU first to reduce FTP writes to NAND. A clean coarse IU is simply dropped from the table and marked spilled because its FTP copy is still valid. If there is no clean coarse IU, DUET picks the coarse IU with the most fine pages, so one FTP program frees the most table slots.

GC. When GC moves fine pages of spilled coarse IUs, it reads and programs each FTP at most once per victim block. A merge of a spilled coarse IU reads its entries from the FTP and then clears its F and S bits.

E. Overhead

The per-coarse-IU state is the 4 B coarse entry with its three flag bits, the fine map is the two-choice hash table, and the GTD adds 4 B per FTP, all within the budget. A host read takes at most three DRAM accesses, and a spill or load touches at most $1024/U$ coarse entries and 1024 table slots. Mapping reads and programs happen only for spilled coarse IUs and are counted in all WA and latency results.

IV. Evaluation

Setup. TODO: NVMeVirt configuration, OP 7%, 1/8 mapping DRAM, the seven FTLs, workloads.

Read latency. TODO (Figure 5).

Write amplification and write latency. TODO (Figure 6).

Throughput. TODO (Figure 7).

Ablation. TODO (Figure 8).

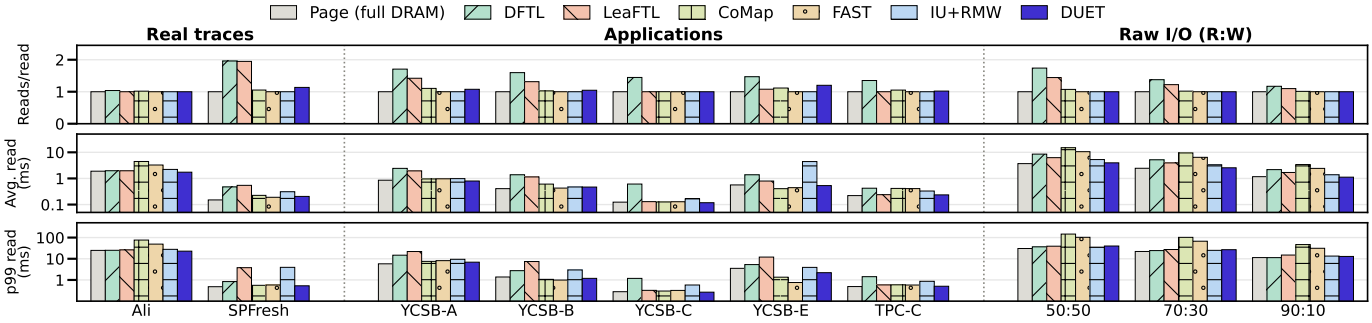


Fig. 5. Read path. From top, flash reads per host read (data plus mapping reads), and average and p99 read latency.

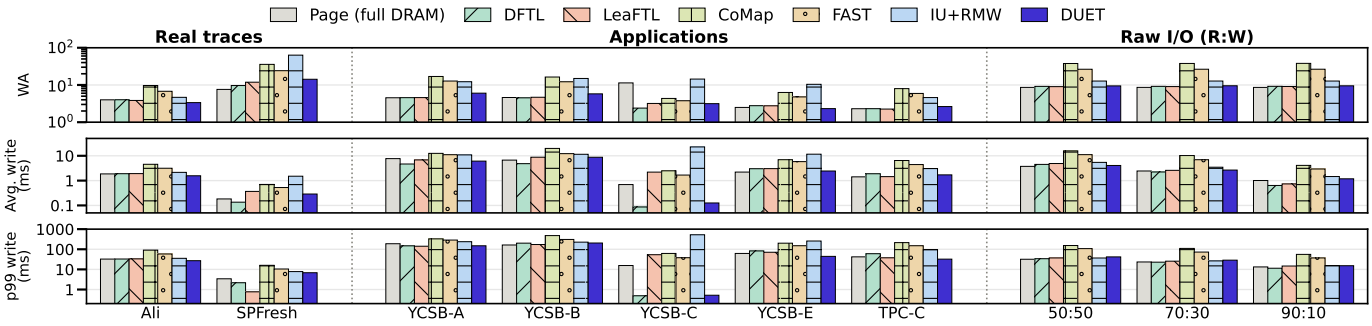


Fig. 6. Write path. From top, device write amplification, and average and p99 write latency.

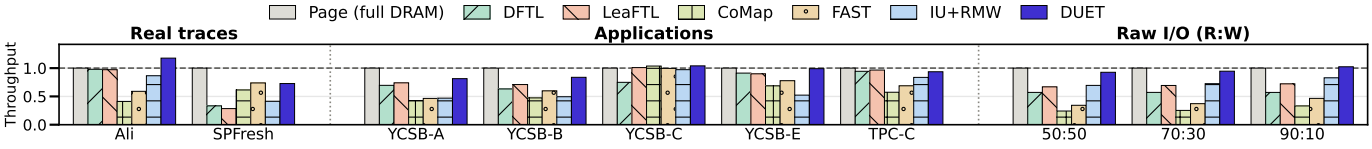


Fig. 7. Throughput normalized to Page.

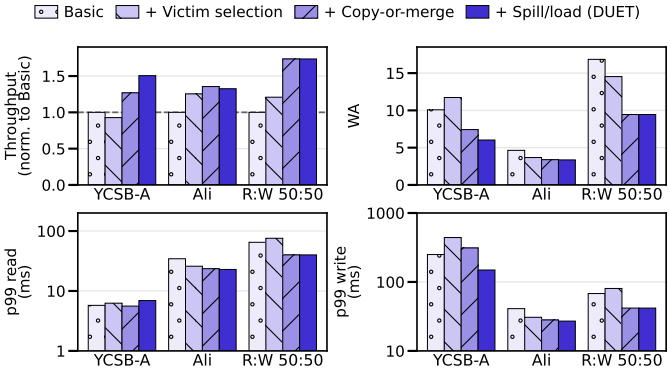


Fig. 8. Ablation. From left to right, each bar adds one more technique to Basic, and the rightmost bar is DUET. Throughput is normalized to Basic.

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